SMRG Flusim Web Dashboard
User Manual

This document will teach you how to use the SMRG Flusim Web Dashboard to run and visualise infectious disease simulations.

Contents

1 Configuring Simulations **3**

1.1 Setting Baseline Parameters 3

1.2 Parameter Types 5

1.3 Additional Scenarios 8

1.4 Other Settings 9

1.4.1 Advanced Parameters 9

1.4.2 Downloading and Uploading Parameters 10

1.4.3 Loading Parameters from Templates 10

2 Running Simulations **12**

2.1 Beginning Simulation Experiments 12

2.2 Simulations in Progress 13

3 Visualising Simulation Results **16**

3.1 Infection-Over-Time Graphs 16

3.2 Health Burden Tables 18

4 Additional Notes **22**

4.1 Simulation Errors 22

4.2 Parameter Errors 23

Glossary **24**

A Advanced Visualisation Options **25**

A.1 Table Configuration 25

A.2 Infection-Over-Time Graph Configuration 28

1 Configuring Simulations

1.1 Setting Baseline Parameters

The SMRG Flusim Web Dashboard allows for many aspects of a simulation to be configured. In order to run a simulation, you must first set these model parameters to values matching the infection you wish to simulate. Click on the “ Baseline Parameters” link on the sidebar (see Figure 1) to view the parameters that can be changed.

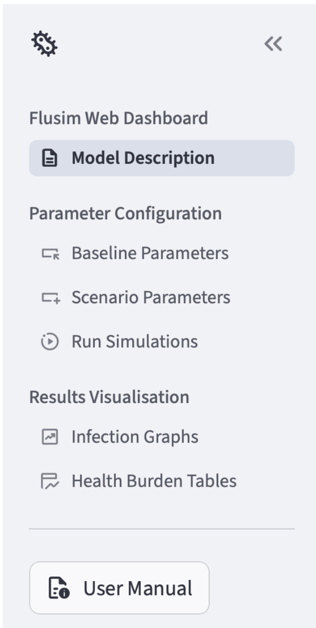


Figure 1: The navigation sidebar. Click on one of the page names to visit that page. If you can’t see this sidebar, click the » arrow on the top-left of the page to make it visible.

The “ Baseline Parameters” page can be used to control the values of the parameters used by the simulation engine. The main method of configuring parameters is through the tabs at the bottom of the page, seen in Figure 2. These tabs separate the simulation’s range of parameters into several categories based on what aspect of the simulation they modify. Click on any of these tabs to view their model parameters.

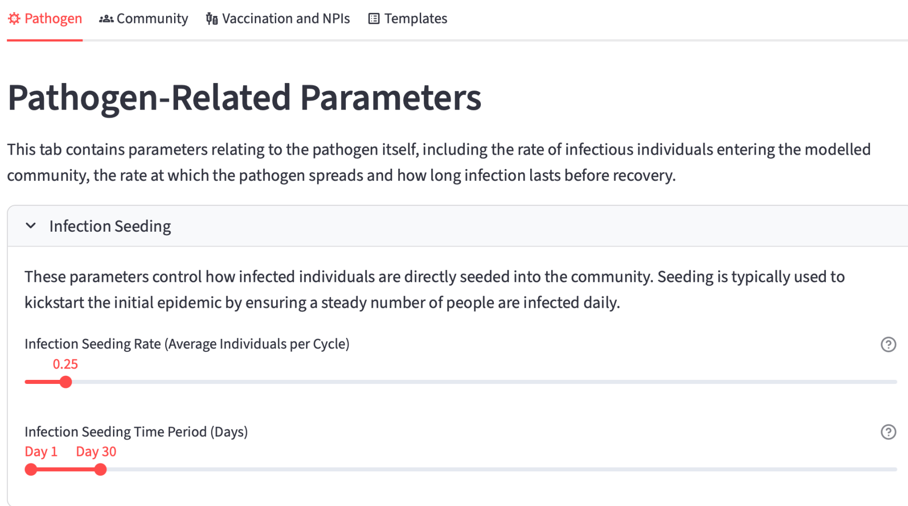


Figure 2: A section of the Pathogen tab on the "Baseline Parameters" page. This section has a drop-down box containing a pair of sliders that control the rate at which infected individuals are seeded into the simulation model. These settings are used to initiate pathogen spread in the modelled community. Click on one of the tab names at the top to switch to that tab. Click the header of the drop-down box to open or close it. Click and drag the circles on the sliders to change the infection seeding settings.

- **Pathogen** relates to the infection itself, including how frequently it infects new individuals and the length of time individuals will be infectious for.
- **Community** controls the simulated population and how they react to the pathogen, including the size of gatherings and how likely a sick individual is to stay at home.
- **Vaccinations and NPIs** controls the impact of vaccination strategies and non-pharmaceutical interventions (techniques used to limit the spread of infection without medication, e.g. social distancing).
- **Dynamic** allows factors such as NPI effectiveness to change at specific times during the simulation period. This tab is not visible by default, but becomes visible when advanced parameters are enabled (see Other Settings).
- **Templates** allows loading entire parameter sets from preset templates. Further information can be found in Loading Parameters from Templates.

In each tab, several parameters may be adjusted. Different parameters use different widgets to control their values; see Parameter Types for an explanation of the different kinds of parameters. If you do not know what a specific parameter does, hover your mouse over the ⓘ symbol at the top-right of that parameter to show a description. Some parameters are enclosed in drop-down boxes like the one in Figure 2; click on a box to open or close it.

1.2 Parameter Types

Below is a list of the input methods used by parameters on the dashboard.

Sliders

Sliders may be used to set most numeric parameters on the dashboard. In addition to clicking and dragging the slider, you can also click a point on the bar to move the slider there immediately, or use the left and right arrow keys to fine-tune the values of sliders.

Some sliders, like the one in Figure 3, have two values on the same line. The two values are the start and end of the range the parameter uses; either one can be adjusted freely.

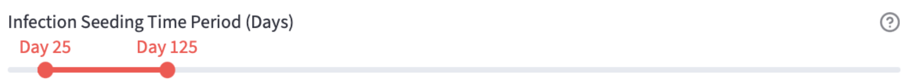


Figure 3: A slider with two values to adjust. This selects a period of time over which infected people will be added to the simulation.

Toggles

Some parameters can either be enabled or disabled in the simulation. Parameters like the one shown in Figure 4 can be toggled on and off by clicking the toggle switch. Note that some parameters are dependent on these toggles; for example, if "Enable Vaccines in Simulation" is toggled off you won't be able to edit other vaccine-related parameters unless you toggle them back on.

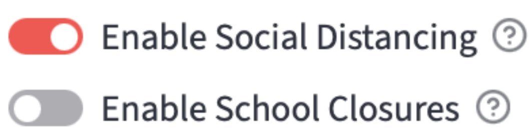


Figure 4: Toggle parameters which control whether specific non-pharmaceutical interventions are utilised in the simulation. The first is enabled, while the second is disabled.

Selections

If there are multiple options for a particular parameter, it will use a selection box like the one in Figure 5. Click on the box to reveal the options, then click on the option you want. Remember that hovering your mouse over the ⓘ symbol will show a description of the parameter, including what each option relates to.

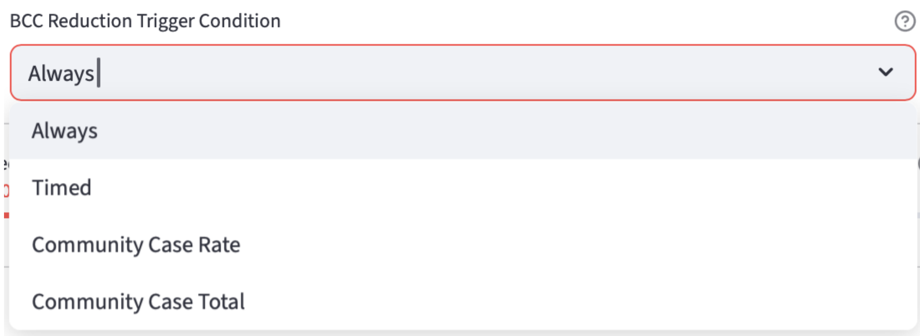


Figure 5: A parameter that uses a selection box to determine what type of condition will enable the background contact count reduction NPI in the simulation.

Number Inputs

Parameters that require more precision than a slider can provide (such as Figure 6) require you to enter a number directly. Simply type the number you want and press Enter to confirm, or click the + and − symbols to increase or decrease the value by a small amount. If you enter an invalid value (e.g. a negative number), the dashboard will prompt you to enter something else.

Figure 6: A number input parameter controlling how many people in the modelled community receive their first vaccination per day.

Tables

Some parameters can be set to a unique value for different age groups (see Figure 7). For parameters like these, a table is used to set the values for each age group. Double-click on a cell in the table to begin editing its value. For "Age Group" columns, select the age group that you would like to define age-specific parameter for using the drop-down menu. For numeric columns, simply type the new value and press Enter on your keyboard; if the column uses percentages, you will need to enter the value as a decimal.

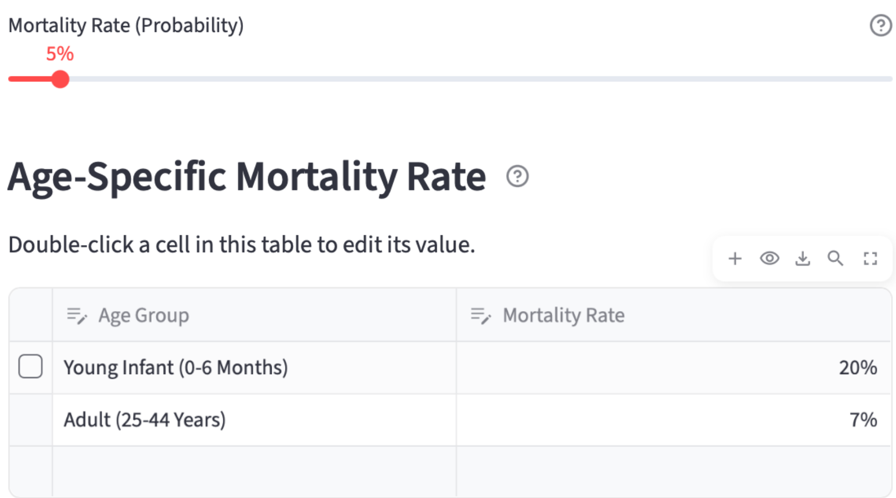


Figure 7: A parameter that uses a table to give different age groups different likelihoods of dying following infection.

You may add or remove rows from these tables freely. To add a row, hover your mouse over the row at the bottom of the table until a + icon appears, then click to add the row. Make sure to select an age group once the row has been added. Each row should use a different age group; if multiple rows have the same age group an error message will appear (see Parameter Errors for more details). To remove a row, click on the check box next to the row in the thin column to the left of the table, then click the delete button in the top-right corner of the table. Note that these buttons will only be visible when your mouse cursor is hovering over their location.

The parameters in the Dynamic tab (see the descriptions of tabs in Setting Baseline Parameters) also use tables for input. Instead of selecting an age group with the left column, you may enter a numbered day of the simulation where a parameter's value will change.

1.3 Additional Scenarios

To compare the effects that different parameter settings have on the simulation, the dashboard allows the definition of multiple parameter settings, or ‘scenarios’. Scenarios typically relate to alternative mitigation strategies involving vaccination and/or non-pharmaceutical interventions, but can be used for any experiments that require comparing the spread of infection under different environments.

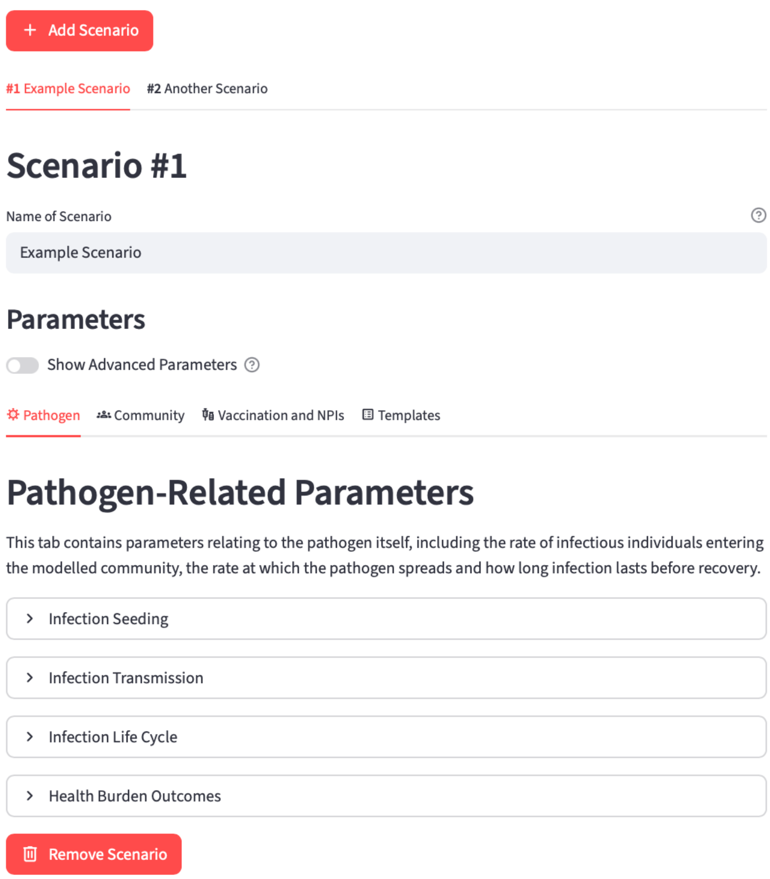


Figure 8: The panel where additional modelling scenarios may be created. Click “+ Add Scenario” to add an additional scenario to the simulation, then change parameters from their baseline values using the parameter tabs.

The parameters set on the “Baseline Parameters” page are the values that will be used by the first scenario, also called the baseline scenario. In order to add additional scenarios, use the sidebar to navigate to the “Scenario Parameters” page. Click the “+ Add Scenario” button seen in Figure 8 to add another scenario to the dashboard; up to 30 extra scenarios may exist at once.

On creating a new scenario, you will be presented with the panel seen in Figure 8. Each scenario is shown as a tab within this panel; click on the name of the scenario to view its parameters. Type a name in the "Name of Scenario" field and press Enter to give the scenario a name. This name will be used on all graphs and tables generated from the output of simulation experiments that include the scenario. Two scenarios cannot have the same name; if you attempt to give a scenario a name that already exists, an error message will appear and the name will not be changed.

The panel may also be used to configure the new scenario's parameters separately from the baseline scenario. This uses the same set of parameter tabs as the baseline scenario, as seen in Figure 8. By default, the new scenario's parameters will be set to the values chosen for the baseline scenario, but they may be changed freely. You may also remove the scenario by clicking the " Remove Scenario" button below a scenario's name.

1.4 Other Settings

In addition to the parameter tabs, there are several other options available on the " Baseline Parameters" and " Scenario Parameters" pages, as seen in Figure 9.

Figure 9: The additional settings for parameter configuration, including the advanced settings toggle, " Download Parameters to File" button and " Upload Parameters from File" button.

1.4.1 Advanced Parameters

Due to the complexity of the *Flusim* model, not all parameters are shown on the dashboard by default. These parameters include separate asymptomatic infection likelihoods for children and adults, waning immunity to the pathogen, and dynamic modification of parameter values at specific times during the simulation period. By enabling the "Show Advanced Parameters" toggle above the parameter tabs, these advanced parameters will become visible in the parameter tabs.

1.4.2 Downloading and Uploading Parameters

Parameters that have been configured on the dashboard do not save between sessions. To avoid needing to reenter parameters when beginning a new session (e.g. after refreshing the page), the parameters that you have configured can be downloaded as a JSON file. Click the " Download Parameters to File", then click "Confirm" on the resulting dialog box to download the parameters.

The parameters for each scenario configured on the dashboard are stored together as a JSON file named `FlusimParameterSettings[timestamp].json`. While the name of the file can be changed freely, it is recommended that you do not modify the file's contents. If the file no longer matches the proper JSON format, you will not be able to load its parameters.

To reload a set of downloaded parameters, click the " Upload Parameters from File" button, then click "Browse files" in the resulting dialog box. Upon selecting a valid parameter JSON file, the dashboard will set all parameters on the " Baseline Parameters" and " Scenario Parameters" pages to match the values in the file.).

> Uploading parameters from a file will overwrite any parameters that are already set on the dashboard, including scenarios that have been added and simulation engine settings (see Beginning Simulation Experiments).

1.4.3 Loading Parameters from Templates

In addition to loading parameters for the entire simulation, you may also load parameters for a single scenario using templates. Navigate to the Templates tab while configuring a scenario's parameters.

In order to load a template, click on the button for the desired template (as seen in Figure 10 and click "Confirm" in the resulting dialogue box. All parameter values in the scenario will be replaced with the values from the selected template, though the scenario's name will not change.

> Loading a template onto the baseline scenario may modify the values used in additional scenarios that inherit from it. It is recommended to check the parameters in other scenarios after updating the baseline scenario this way to ensure that the desired parameters are kept.

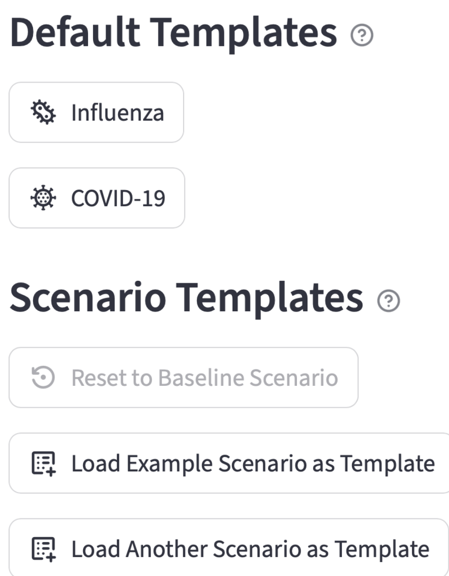


Figure 10: The templates visible when accessing the Templates tab for the baseline scenario. In addition to the two preset templates, there are buttons to load two additional scenarios (named "Example Scenario" and "Another Scenario") as templates, and a button to reset parameters to their baseline values (which is greyed out due to this already being the baseline).

By default, two preset templates are available to load:

- **Influenza** uses parameters that simulate a simple influenza pandemic. In addition to using transmission probabilities that match influenza's rate of infection, this preset also includes initial vaccination proportions that emulate Australia's influenza vaccine coverage.
- **COVID-19** uses parameters that simulate an outbreak of SARS-CoV-2 Delta. This preset also enables background contact count and withdrawal increase non-pharmaceutical interventions, simulating a realistic response to such an outbreak.

Additionally, if additional scenarios have been defined for the simulation, you may choose to copy their parameters as a template for other scenarios, as seen in Figure 10. This makes it easy to make multiple scenarios with similar changes from the baseline. Using the baseline scenario as a template will reset all of a scenario's values back to the values defined in the baseline scenario.

2 Running Simulations

2.1 Beginning Simulation Experiments

To run scenarios as part of a simulation experiment, use the sidebar to navigate to the "Run Simulations" page. This page (depicted in Figure 11) contains several parameters which control the settings of the simulation engine. These settings control aspects like how long the simulation will run for (in days), how many simulation runs to perform with each scenario (to account for stochasticity) and what community the model will be based on. Simulation engine settings are applied to all scenarios rather than being individually configurable.

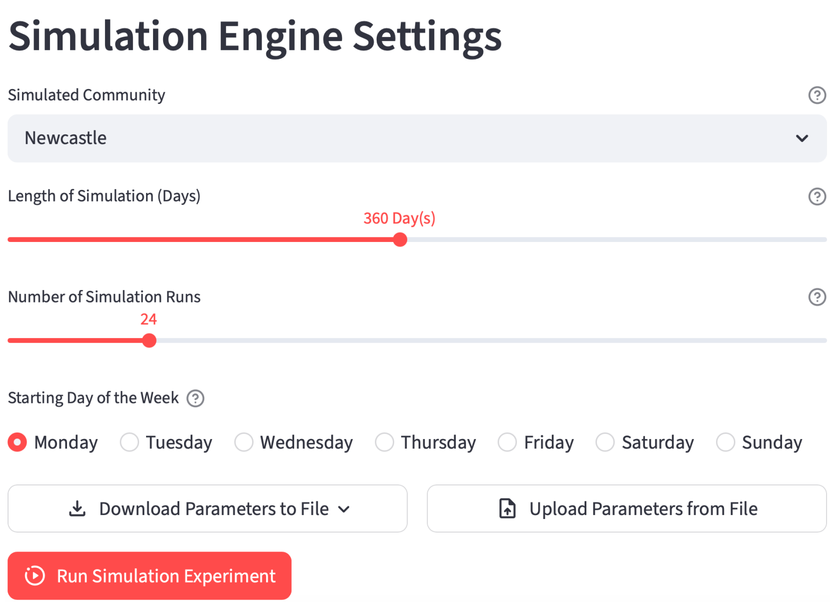


Figure 11: The "Run Simulations" page, with settings to configure the simulation engine and the parameter download/upload buttons seen in Other Settings. Click the "Run Simulations" button below them to run your scenarios together.

Once the simulation engine settings are configured, click the "Run Simulations" button. The confirmation prompt seen in Figure 12 will appear on the screen, listing the names of all configured scenarios. The dashboard will also provide an estimate for how long it will take to run the simulation experiment, based on the number of scenarios and the simulation engine settings. If there are any errors that result from the chosen parameter settings, such as an NPI being set to begin after the simulation has already ended, they will be listed beneath the parameter names (see Parameter Errors for more details). Click "Confirm" at the bottom of the prompt to run the simulation experiment.

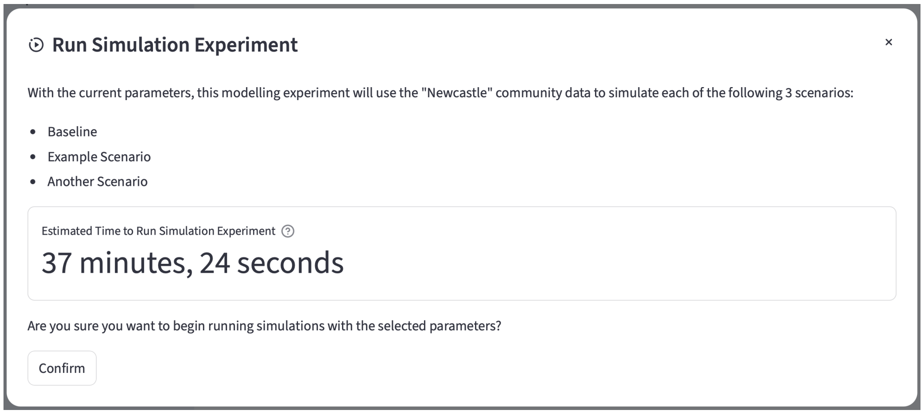


Figure 12: An example of the confirmation prompt that will appear when you attempt to run the simulation. In this instance, the user has added 2 additional scenarios, naming them "Example Scenario" and "Another Scenario". Running these scenarios is estimated to take 37 minutes and 44 seconds. Click "Confirm" to send the parameters to the simulation engine.

2.2 Simulations in Progress

When you run a simulation experiment, the dashboard sends the parameters for your scenarios to a separate server. This server will use these parameters to run the infectious disease model using the simulation engine.

Due to the use of stochastic parameters (such as the probability of transmission between two individuals), a single simulation experiment will involve multiple runs of the simulation engine for each scenario. Once all simulations are complete, the server will use the outcomes of each simulation run to calculate the median daily infection counts for each scenario, then send these results back to the user dashboard. While this process is occurring, you will not be able to run any new scenarios, but you may still navigate to other pages on the dashboard.

> Simulation experiments can take a long time to complete if you have defined many scenarios or make each run last a long time. While the estimated time seen in Figure 12 accounts for several of the factors that affect the completion time, additional factors such as the total number of infections that occur in each simulation or the simulation server being busy with other tasks may cause the experiment to take longer or shorter than this estimate.

While the simulation is running, the “Run Simulations” page will update to include the progress bar shown in Figure 13. This progress bar updates to indicate which step of the experiment the server is currently performing. Click on the drop-down container below the progress bar to see a list of all tasks that have been completed so far. If any errors occur when running the simulation experiment, they will be displayed in this task log as well (see Simulation Errors for more details).

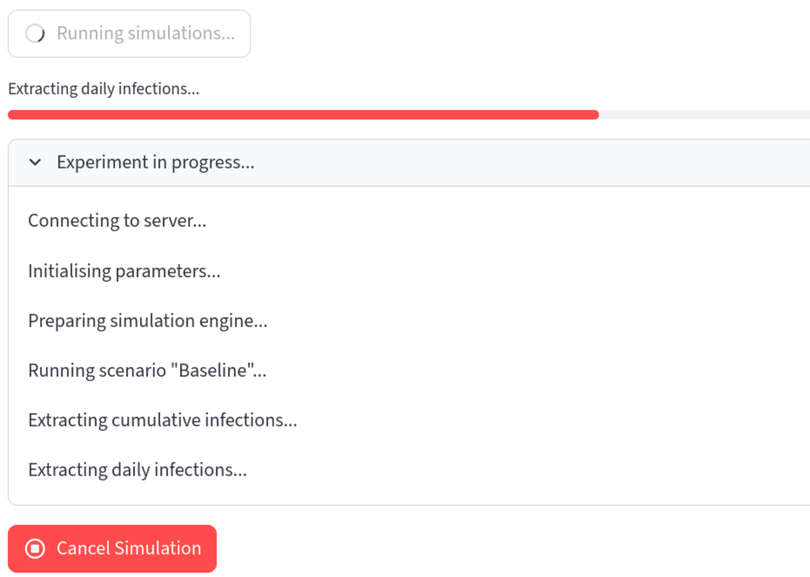


Figure 13: The progress bar displayed when a simulation experiment is running. The task log below the progress bar shows that the server has already ran the baseline scenario used in this experiment and is now extracting age-based infections in order to display them in health burden outcome tables. The “Cancel Simulation” button below the task log can be used to cancel the simulation.

When the simulation is complete, a popup like the one in Figure 14 will appear in the top-right corner of the webpage to inform you that the results are ready to be viewed. You may begin a new simulation by clicking the “Run Simulations” button again after modifying the parameters used by the simulation.

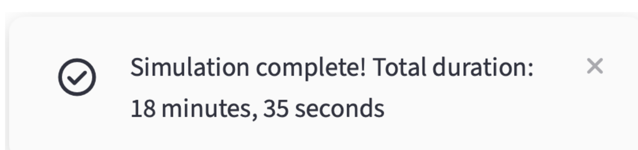


Figure 14: The popup that appears when simulations are complete, showing how long it ran for.

> Do not refresh the webpage after running a simulation. This will reset the session associated with the dashboard, resulting in all parameter settings and simulation results being lost. If you need to refresh the page for any reason, it is recommended that you download your parameters (see Other Settings) and any generated graphs/tables before doing so.

If you wish to cancel a simulation that is in-progress, click the "⏹ Cancel Simulation" seen in Figure 13 and click "Confirm" in the resulting dialog box. This will immediately halt the simulation, allowing you to begin running a new simulation with different parameter settings immediately without waiting for the previous simulation to complete.

3 Visualising Simulation Results

Once a simulation experiment has finished running and the server has returned its results, you will be able to use these results to generate graphs and tables.

3.1 Infection-Over-Time Graphs

The simplest of the visualisations the dashboard can create is a line graph plotting median infections over time (averaged over the multiple simulation runs in each experiment), such as the classical epidemic curves displayed in Figure 15. Use the sidebar to navigate to the "Infection Over Time Graphs" page in order to generate these graphs.

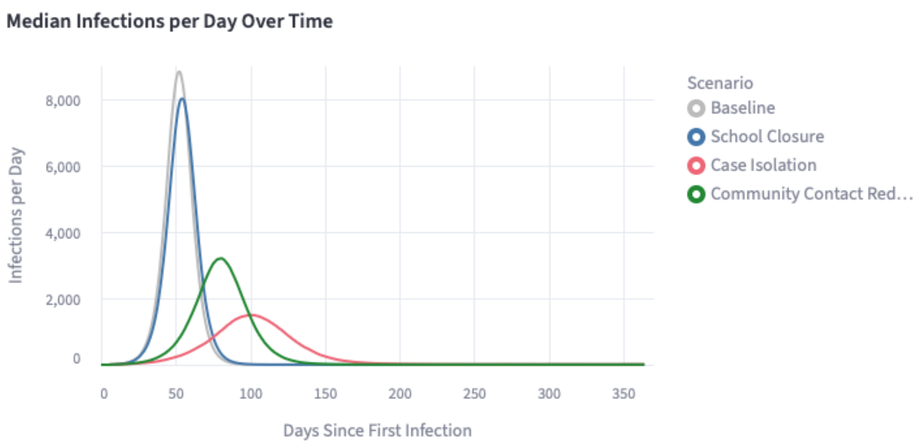


Figure 15: An example of an infection-over-time graph that can be generated using the dashboard, plotting daily infection curves under alternative mitigation measures.

Clicking on the "Graph Settings" menu seen in Figure 16 lets you configure whether the graph should display daily infections or total (cumulative) infections. It also contains a multiple selection box controlling which scenarios are displayed on the graph; by default, all scenarios defined in the "Scenario Parameters" page are included. Click the × next to a scenario to remove it, or click the ⊗ on the right to remove all of them; click the box itself to display any scenarios that can be added.

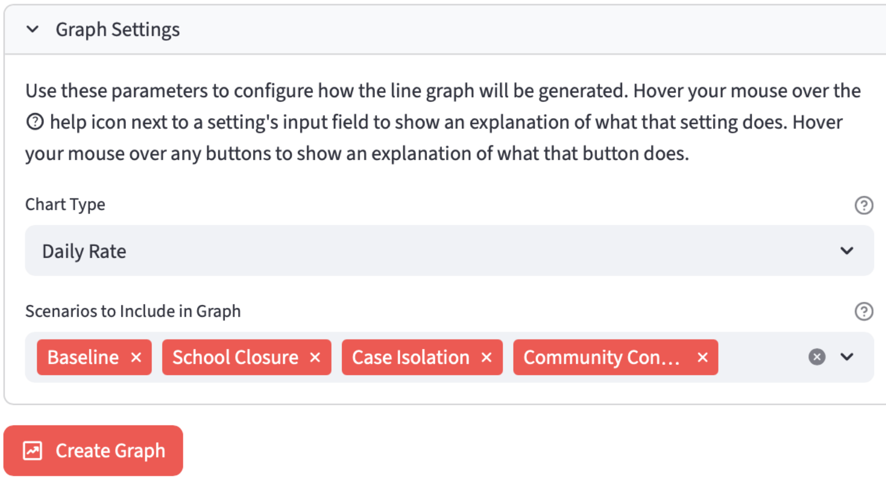


Figure 16: The settings panel for infection-over-time graphs, allowing you to choose the type of graph and which scenarios to include. Click the " Create Graph" button to use these settings to generate a graph.

Click the " Create Graph" button to generate the graph. If the " Create Graph" button is greyed out, it may mean that your simulation hasn't finished running yet; wait until it is complete before trying to generate a graph.

> Occasionally, you may notice that infection-over-time graphs do not extend to the total simulation length configured in the simulation engine settings. This happens when the simulation reaches a point where no new infections occur each day, causing it to end early. Usually this indicates that your transmission probability is either too high or too low; try adjusting its value if you encounter this behaviour.

Infection-over-time graphs are displayed beneath the "Graph Settings" menu. To download a copy of the data used to generate the graph, click the " Download Infection Data" button. Further options for viewing the graph can be seen in Appendix A under Infection-Over-Time Graph Configuration.

3.2 Health Burden Tables

The dashboard may be used to create health outcome tables from the simulation experiment's generated data, compiling health burden outcomes as in Figure 18. These tables display the resulting data as metrics such as diagnosed cases, hospitalisations and deaths. The figures for these tables are made by combining the age-specific infection data extracted from the simulation output with probabilities defining the occurrence of health burden outcomes relative to infections. These probabilities are defined in the Pathogen tab when configuring the parameter settings for the experiment. Navigate to the " Health Burden Tables" page using the sidebar to create these tables.

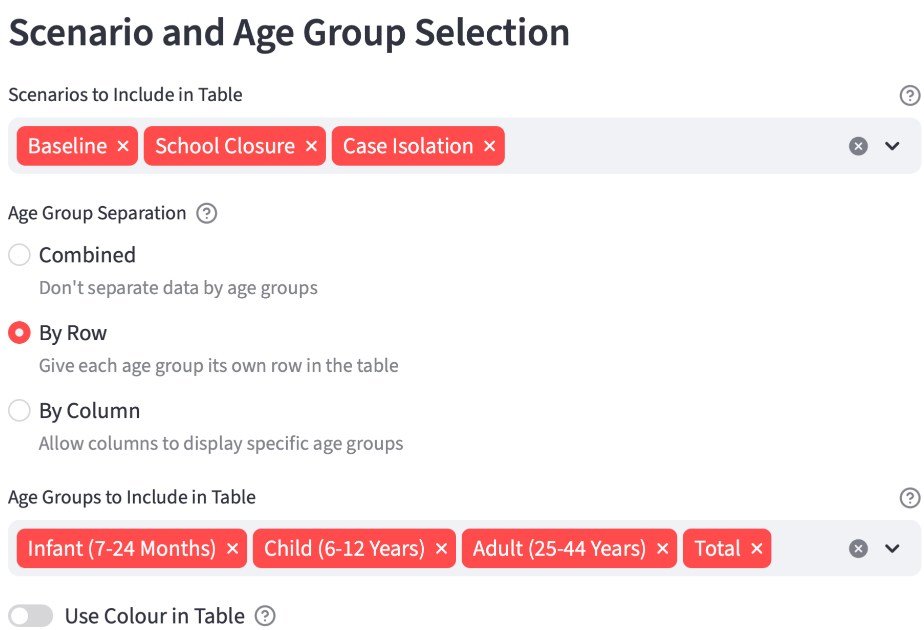


Figure 17: The settings panel for health burden tables, allowing you to configure which scenarios to include and how age groups should be handled. The table where columns can be configured is not pictured here; see Figure 19 to view these column settings.

Tables use a more advanced settings menu than graphs, as in Figure 17.

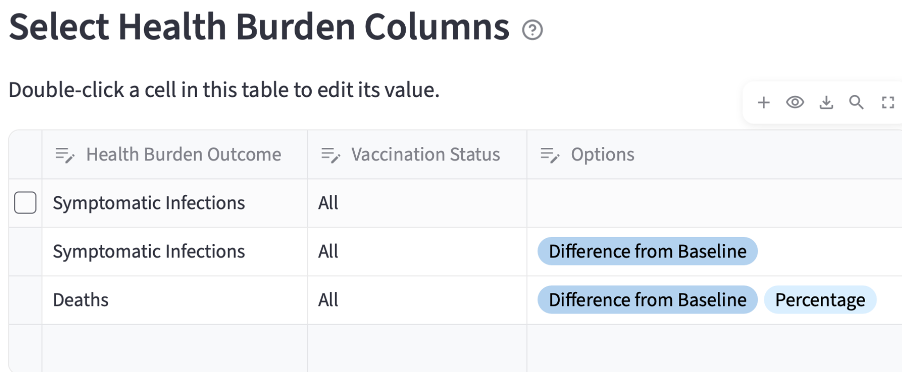
- Scenarios may be added or removed from the table, just as with infection-over-time graphs. Click the × beside each scenario to remove it, or click the ⊗ on the right to remove all of them. Click the selection box itself to display all the scenarios that can be added.

- The “Age Group Separation” setting allows you to control how different age groups are displayed in the final table. Click on any of the 3 options to select it:
  - “Combined” will combine all age groups in the final table, displaying health burdens for each scenario as a whole.
  - “By Row” will make different rows of the table represent different age groups. Each row of the final table will display the health burden outcomes for a specific age group in a specific scenario, as seen in Figure 18. Enabling this option will also reveal the “Age Groups to Include in Table” setting, letting you choose which age groups are given rows in the final table.
  - “By Column” will make different columns of the table represent different age groups. Enabling this will add an additional option to the column settings, allowing you to select which age groups that column should include.
- The “Use Colour in Table” toggle controls whether cells in the final table should be coloured. When enabled, the Scenario column will use a different colour for each scenario; if “Age Group Separation” is set to “By Row”, the “Age Group” column will do the same. Additionally, any columns set to display differences from baseline will colour cells based on the size of the differences. Larger reductions in health burdens will be coloured with a deeper shade of blue, while larger increases will use a deeper shade of red. Figure 18 shows an example of a table with coloured cells.

Scenario	Age Group	Infections	Deaths (Difference from Baseline)
School Closure	Infant (7-24 Months)	4044	-6
School Closure	Child (6-12 Years)	11693	-38
School Closure	Adult (25-44 Years)	50280	-70
Case Isolation	Infant (7-24 Months)	1990	-109
Case Isolation	Child (6-12 Years)	5932	-327
Case Isolation	Adult (25-44 Years)	24942	-1337
Community Contact Reduction	Infant (7-24 Months)	2595	-78
Community Contact Reduction	Child (6-12 Years)	8216	-212
Community Contact Reduction	Adult (25-44 Years)	36543	-757

Figure 18: A health burden table generated using the dashboard listing diagnosed cases between alternative mitigation methods for different age groups. This table highlights how different interventions reduce deaths in different age groups.

The columns that will be included in the table can be configured individually using the table seen in Figure 19, which works similarly to the tables used for age-specific parameters (see Tables).



Select Health Burden Columns

Double-click a cell in this table to edit its value.

	Health Burden Outcome	Vaccination Status	Options
☐	Symptomatic Infections	All	
	Symptomatic Infections	All	Difference from Baseline
	Deaths	All	Difference from Baseline, Percentage

Figure 19: The column selection form for health burden tables.

Double-click on a cell to begin editing it. Each row in the settings table represents a column that will be included in the final health burden table; rows can be added by clicking on the + icons on the bottom row of the table, or removed by clicking the ☐ checkbox on the left side of the table then the delete icon at the top-right. Note that you may need to hover your mouse over the table to see these icons.

The options that can be configured using this table are as follows:

- "Health Burden Outcome" lets you select which health burden the column should display. These health burdens are calculated using the rates defined under the "Health Burden Outcomes" section of the Pathogen parameter tab (see Configuring Simulations); this allows different scenarios to model infections with different levels of severity.
- "Age Groups" appears only when "Age Group Separation" is set to "By Column". Choosing an age group from this list will cause the column to only display health burdens occurring in that age group. Multiple age groups can be selected to combine the health burden values in these groups.
- "Vaccination Status" appears only when vaccines were enabled in the simulation. By selecting "Vaccinated", the column will only display health burdens in individuals who received vaccines in the simulation; the opposite behaviour will occur if "Unvaccinated" is selected. Selecting "All" will display health burdens from both vaccinated and unvaccinated individuals.

- "Options" lets you select options that modify how the columns are formatted. This column allows for multiple selections per row; you may select as many options as you like for each column, or none at all. The options that are available for each column are as follows:
  - The "Difference from Baseline" option makes the column display the difference in health burdens between a given scenario and the baseline scenario, rather than displaying the raw values.
  - The "Percentage" option makes the column display the percentage of the population instead of the raw values.
  - If both "Difference from Baseline" and "Percentage" are selected, the column will display the percentage difference in health burdens between a given scenario and the baseline scenario.

> If the current simulation has waning immunity enabled, columns set to display the percentage of the population may be inaccurate due to the possibility of reinfection. The dashboard will display a warning message if these columns exist for a simulation with waning immunity enabled.

Once you have selected your chosen settings, click the " Create Table" button to generate the table. Like the " Create Graph" button, the " Create Table" button may be greyed out if you have yet to complete a simulation or if there is an issue with the chosen settings. Hover your mouse over the button while it is greyed out to show a tooltip explaining the issue.

To download a copy of the generated table, click the " Download Table Data" button. Further options for viewing the table can be seen in Appendix A under Table Configuration.

4 Additional Notes

This section contains notes on what to do if errors occur alongside other features of the dashboard.

4.1 Simulation Errors

If any issues occur when attempting to run a simulation experiment, the experiment will stop and the popup seen in Figure 20 will appear in the top-right corner of the dashboard. When this occurs, the "Run Simulations" page will update to provide more details about this error. Navigate to this page if you need to restart the simulation or learn more about why the error occurred.

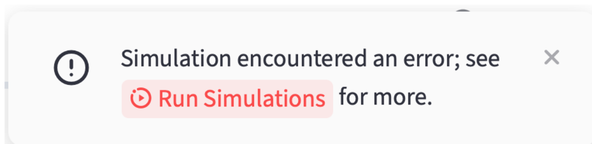


Figure 20: The popup that appears if an error occurs while the dashboard is attempting to run a simulation experiment.

The "Run Simulations" page will display the error on the progress bar and task log shown in Figure 21 (see Simulations in Progress for more details). The progress bar will describe the error that occurred, while the task log will explain what happened in detail and provide potential debugging steps to solve the issue. The full traceback for the error is also included in this log to assist in debugging.

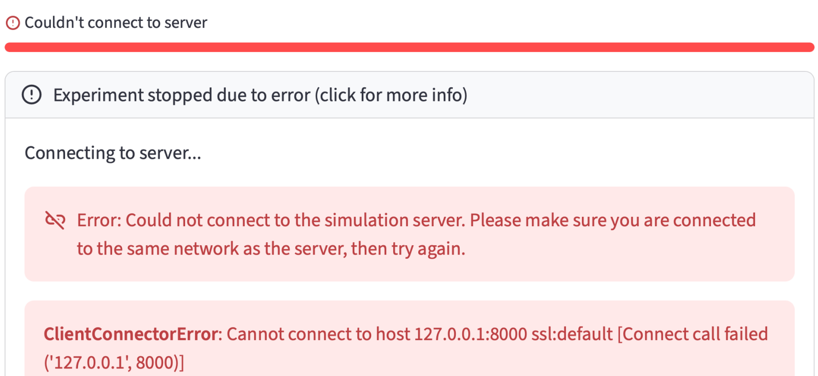


Figure 21: The progress bar and task log seen on the "Run Simulations" page, highlighting an error that occurred while attempting to run the simulation. In this instance, the dashboard could not connect to the simulation server, suggesting that it may have not been on the correct network.

4.2 Parameter Errors

If you set a parameter to a value that would cause issues in the simulation, an error message will appear below that parameter. The error message will explain why the current values cause issues and describe how you can change the parameters to fix the issue. Additionally, each scenario has a drop-down box that contains copies of all the error messages resulting from its parameters. If any errors are present, this box is shown above the parameter tabs for that scenario as well as on the dialog box that appears when running the simulation, allowing you to check if there are any issues with the current parameters at a glance. Figure 22 shows an example of this drop-down box.

Error messages may be either yellow or red. Yellow error messages are warnings; they appear if parameters will cause unrealistic or unintuitive behaviour. You may still run a simulation if your parameters receive yellow error messages, but you should read them carefully as they are likely to result from an oversight when setting parameters. Red error messages are crucial errors; the simulation will not run if any of these are present. Both kinds of error can be seen in Figure 22.

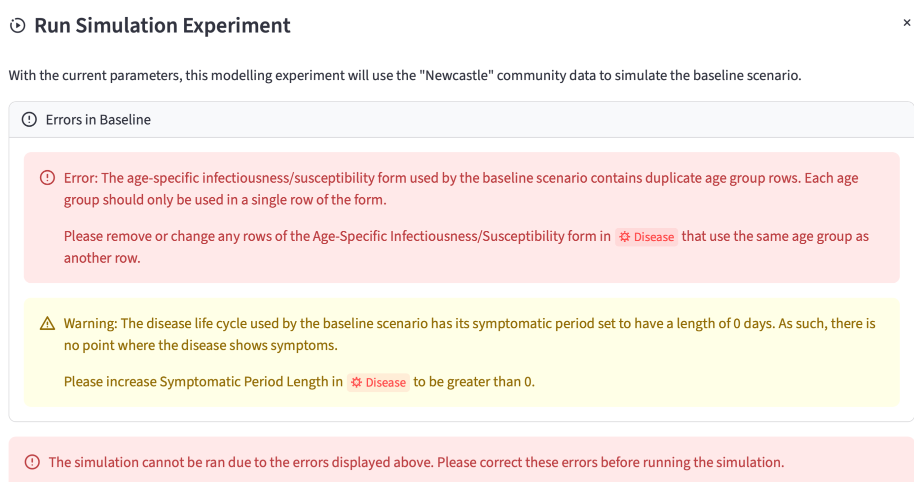


Figure 22: The dialog box for running the simulation, showing the drop-down box that displays the errors present in the baseline scenario. The red error indicates the presence of duplicate parameters that will prevent the simulation from running. The yellow error highlights the infection's lack of a symptomatic period, which may result in unintended behaviour. The "Confirm" button is not present due to the red error stopping the simulation from being ran.

Glossary

***Flusim* model** The agent-based simulation model used by this dashboard to simulate infectious respiratory diseases. 9, 24

background contact count A parameter defined as part of the *Flusim* model, which determines how many interactions between random people in the simulation occur each day. It is used to emulate interactions that occur outside of the model's simulated locations, such as on public transport or in sporting events. 6, 11

baseline scenario The scenario defined in the " Baseline Parameters" page. All other scenarios within a simulation experiment will use the parameters defined in this scenario as default values for parameters that the scenarios do not have their own values for. 8, 10, 11, 14, 21

health burden outcome A statistic measuring a negative consequence of an infectious disease outbreak. Examples include GP visits, hospitalisations and deaths. 14, 18, 24

non-pharmaceutical intervention Any intervention or protocol implemented to combat the spread of disease that does not involve medication or pharmaceutical products. Examples include social distancing and school closure. 4–6, 8, 11, 12

scenario A set of parameters defining a single environment for the simulation. 8, 10, 12, 13, 16–18, 24

simulation engine The program that conducts simulations according to the *Flusim* model, hosted separately from the dashboard itself. 3, 10, 12, 13, 24

simulation experiment The process that is triggered upon clicking the " Run Simulations" button, in which multiple scenarios are ran together by the simulation engine in order to compare infection rates and health burden outcomes between them. 9, 12, 14, 16, 18, 22, 24

simulation run The act of using the simulation engine to simulate a given scenario a single time. Simulation experiments use multiple runs for each scenario to account for the presence of stochastic parameters. 12, 16

stochasticity The property of being random or unpredictable. Several parameters used by the *Flusim* model are stochastic in order to replicate the unpredictable nature of infectious diseases. 12, 13, 24

A Advanced Visualisation Options

As part of the Streamlit framework used to construct the dashboard, the graphs and tables throughout the dashboard have many interactive options to change how they are displayed. This appendix will list all of these options.

A.1 Table Configuration

Tables, like the ones used to configure age-specific parameters (see Tables) or the health burden tables that can be generated using the results of simulation experiments (see Health Burden Tables) utilise many forms of interactivity. Note that many of the buttons and icons mentioned here are not visible unless your mouse is hovered over the table.

- If the generated table is too large for the dashboard to display at once, you can scroll the table both vertically and horizontally using the scroll bars that appear on the right and bottom edges. The scroll wheel on an optical mouse will also work; holding Shift will allow you to scroll horizontally with the scroll wheel instead of vertically.
- The Scenario and Age Group columns in health burden tables use unique colours for each scenario/age group in the table. Additionally, any "Difference from Baseline" columns are colour-coded based on the size of the difference; greater decreases in health burden outcomes are more blue, while greater increases are more red.
- Clicking on a cell will highlight it. Clicking and dragging allows multiple cells to be highlighted. When cells are highlighted, the standard copy hotkeys (ctrl+C, or ⌘+C on Mac computers) can be used to copy the values in the highlighted cells to your clipboard.
- Double-clicking on a cell containing a number will display the exact value of that cell, without any rounding or other formatting.
- Hovering your mouse over a column header will show a tooltip explaining what the values in the column represent.
- Clicking on a column header will sort the table using that column, in ascending order. Clicking again will switch to descending order, and clicking a third time will remove the sorting. Note that only one column can be used for sorting at a time; clicking on another column header will remove the current sorting.
- Clicking and dragging a column header left or right allows you to reorder the columns in the table.

- Clicking and dragging the borders between column headers allows you to adjust how wide each column is.
- Clicking on the ⋮ menu dots to the right of a column header will open a drop-down menu where different settings for the column can be modified, seen in Figure A1. From here, the following options are available:

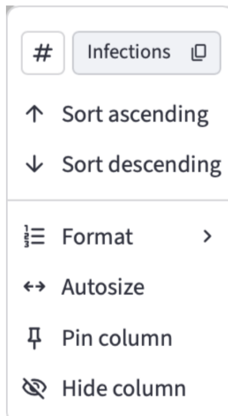


Figure A1: An example of the list of options that can be used to configure a column's appearance within a table.

  - Clicking the copy symbol next to the column's name will copy the name of the column to the clipboard.
  - The ↑ Sort ascending and ↓ Sort descending options will sort the table in the same way as clicking on the header does.
  - The Format option on numeric columns allows you to change how the column's numbers are formatted. The available options include 1.75 Plain (no formatting or rounding), Compact (uses suffixes like K for 1000 to abbreviate numbers), % Percent (displays values as percentages rounded to 2 decimal places) and Scientific (displays values using scientific notation, e.g. 1.873e5), among other, more specialised options.
  - The ↔ Autosize option will adjust the width of the column so that it is exactly enough to display the column header and all currently-visible cells without cutting anything off.
  - The Pin column option will force the column to always appear on the left edge of the table, even if the table is horizontally scrolled to the right. The Scenario and Age Group columns in health burden tables are pinned by default; the options for these columns will include an Unpin column option to disable this.
  - The Hide column option will remove the column from the table.

- The top-right corner of a table includes settings to configure the table as a whole. The following options can be selected here:
  - The eye icon opens a menu where columns can be added or removed from the table. Columns hidden from their own menu can be re-added here; click the checkboxes next to each column name to show or hide the column. Most tables also have an index column that is hidden by default, but can be added through this menu.
  - The download icon will download a CSV copy of the table. Note that sorting, removing or reordering columns will affect the CSV version of the table as well.
  - The search icon will open up a panel allowing you to search for specific values. Typing a specific value and pressing Enter will highlight cells that contain that value. From here, you may use the ↑ ↓ arrow icons to jump between these cells.
  - The fullscreen icon will expand the table to fill the entire webpage. To return the table to normal size, use the exit fullscreen icon.
  - In tables used to configure settings, the + add icon will add a blank row to the bottom of the table. If any rows have been selected using the checkboxes on the left edge of the table, this icon will change to a delete icon, which will remove the selected rows from the table.

A.2 Infection-Over-Time Graph Configuration

Like tables, infection-over-time graphs incorporate several interactive features.

- Hovering your mouse over the graph will display a tooltip showing the values for each scenario at that time-point on the graph.
- Clicking on the name of a scenario to the right of the graph will hide the lines for all scenarios except that one. Hold Shift while clicking on another scenario name to make that scenario's line reappear without hiding other visible scenarios.
- Clicking the fullscreen icon in the top-right corner will expand the image to fill the entire webpage. To return the image to normal size, click the exit fullscreen icon.
- Clicking the chart icon in the top-right corner will hide the graph and display a spreadsheet containing the data used to generate the graph. This spreadsheet uses many of the same options for formatting that other tables on the dashboard use (see Table Configuration). To return to displaying the graph, click the graph icon.
- Clicking the ⋯ menu dots will display a list of miscellaneous options. The Save as SVG and Save as PNG options allow the graph to be downloaded as an image in either file format. The other options allow you to view the Vega code that is used to generate the graph.